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SYLLABUS

Module I:

MODULE I

Elemental and compound semiconductors, Intrinsic and Extrinsic semiconductors, concept of effective mass, Fermions-Fermi Dirac distribution, Fermi level, Doping & Energy band diagram, Equilibrium and steady state conditions, Density of states & Effective density of states, Equilibrium concentration of electrons and holes.

Excess carriers in semiconductors: Generation and recombination mechanisms of excess carriers, quasi Fermi levels.

PART A

1 Define Fermi-Dirac distribution function of semiconductor. (3)

2 Draw the energy band diagrams under equilibrium for i) Intrinsic ii) n-type iii) p-type semiconductors. (3)

Pause

1)

$f(E)$ specifies, under equilibrium conditions, the probability that an available state at an energy E will be occupied by an electron. It is a probability distribution function.

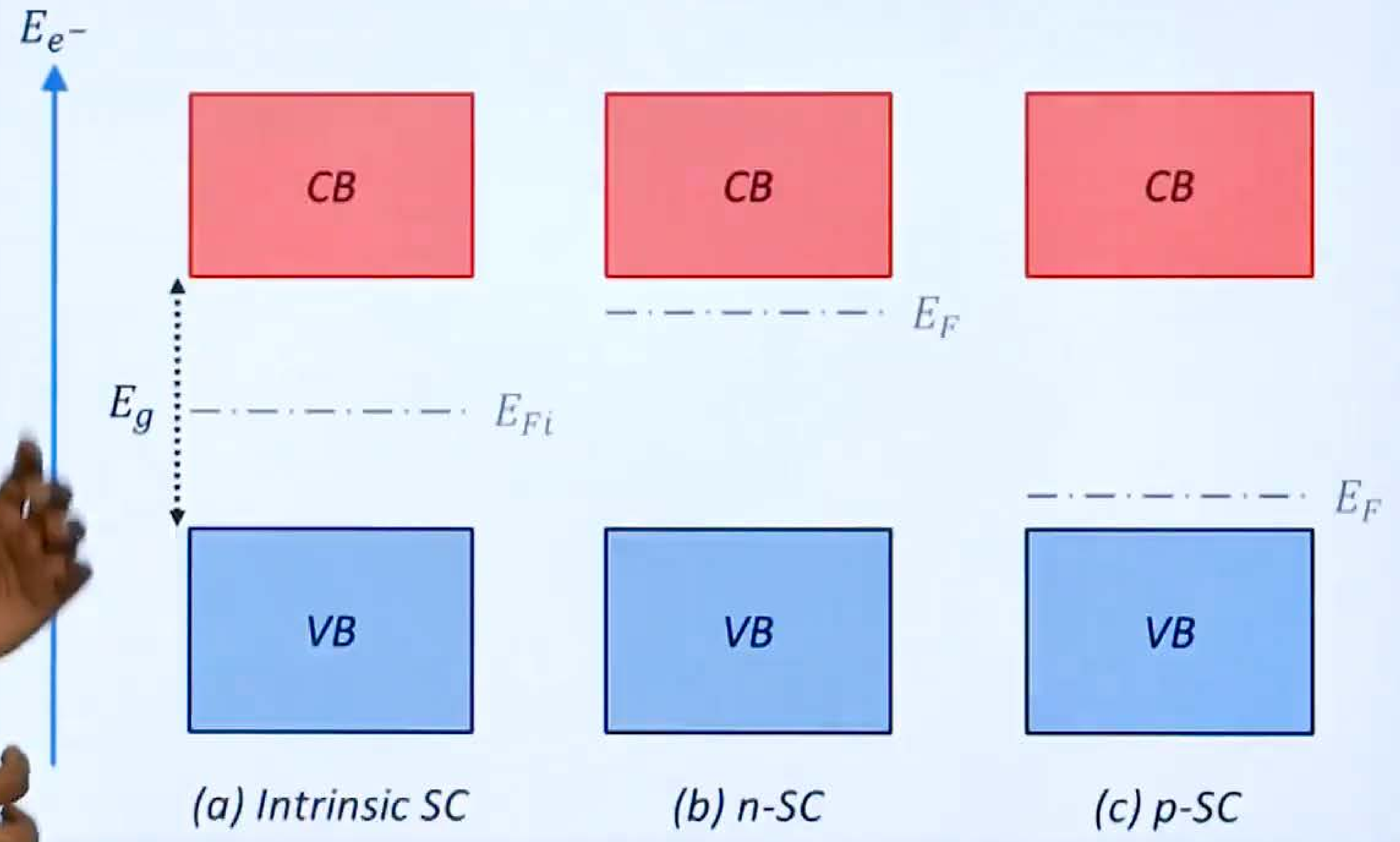
$$\boxed{f(E) = \frac{1}{1 + e^{(E-E_F)/kT}}} \quad (2.7)$$

E_F = Fermi energy or Fermi level

k = Boltzmann constant = 1.38×10^{-23} J/K
= 8.6×10^{-5} eV/K

T = absolute temperature in K

2)



PART B

Module 1

Derive the expression for electron concentration (n_0) and hole concentration (p_0) at equilibrium. (8)

(b) For a silicon sample at 300K, the equilibrium hole concentration is 4×10^{12} cm^{-3}. Determine (i) equilibrium electron concentration (ii) the acceptor concentration if the donor concentration is 10^{12} cm^{-3}. (Assume ni for silicon is 1.5×10^{10} cm^{-3}). (6)

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11 a)

$$n_0 = \int_{E_c}^{\infty} f(E)\,N(E)\,dE$$

∞

E_c

$$\boxed{n_0 = f(E_c)\,N_c}$$

E_c N_c

$$f(E_c) = \frac{1}{\cancel{1} + e^{(E_c - E_F)/kT}} \approx e^{-(E_c - E_F)/kT}$$

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$$n_0 = N_c e^{-(E_c - E_F)/kT} \Rightarrow E_c - E_F = kT \ln\left(\frac{N_c}{n_0}\right)$$

$f(E_c)$

The effective density of states $N_c = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}$

$$n_0 = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2} e^{-(E_c - E_F)/kT}$$

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By similar arguments, the concentration of holes in the valence band is

$$p_0 = N_v \left[1 - f\left(E_v\right) \right]$$

where N_c is the effective density of states in the valence band.

The probability of finding an empty state at E_v

$$1 - f\left(E_v\right) = 1 - \frac{1}{1 + e^{(E_v - E_F)/kT}} \approx e^{-(E_F - E_v)/kT}$$

$$\boxed{p_0 = N_v e^{-(E_F - E_v)/kT} \Rightarrow E_F - E_v = kT \ln\left(\frac{N_v}{p_0}\right)}$$

$$N_v = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$$

$$\boxed{P_o = 2\left(\frac{2\pi m_p^* KT}{h^2}\right)^{3/2} e^{-(E_F - E_v)/KT}}$$

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PART B

Module 1

11 (a) Derive the expression for electron concentration (n_0) and hole concentration (p_0) at equilibrium. (8)

(b) For a silicon sample at 300K, the equilibrium hole concentration is 4×10^{12} cm^{-3}. Determine (i) equilibrium electron concentration (ii) the acceptor concentration if the donor concentration is 10^{12} cm^{-3}. (Assume ni for silicon is 1.5×10^{10} cm^{-3}). (6)

$P_0 \rightarrow$

$n_0 = ?$

$N_A = ?$

$N_D \rightarrow$

7/40

11 b) i) $P_0 = 4 \times 10^{12}$

$n_i = 1.5 \times 10^{10}$

$n_0 = ?$

$n_0 = 5.62 \times 10^{7}$

$n_0 P_0 = n_i^2$

$n_0 = n_i^2 / P_0 = \frac{(1.5 \times 10^{10})^2}{4 \times 10^{12}} = \frac{2.25 \times 10^{20}}{4 \times 10^{12}}$

10^8

$N_D = 10^{12}$

$N_A = ?$

$= 5 \times 10^{12}$

$P_0 + N_D = n_0 + N_A$

$N_A = P_0 + N_D - n_0$

$= 4 \times 10^{12} + 10^{12} - 5.62 \times 10^{7}$

PART B

[illegible]) Explain different types of recombination mechanisms. (7)

[illegible]) A silicon sample doped with $10^{16}cm^{-3}$ donors at 300K is optically excited such that the optical generation rate is 10^{20} EHP/(cm^{-3} s^{-1}). Find the separ[illegible]ion between Quasi Fermi levels and show the positions of equilibrium and [illegible] ermi levels if $\tau_p = \tau_n = 2\ \mu s$. (7)

12 a)

tive Recombination

ative reco tion occurs when an electron from the conduction band recombines with a ole in the band. As a result of electron-hole recombination, a photon is emitted. adiative rec nation is the radiative transition of the electron from the conduction band to valence , which involves optical processes: spontaneous emission, absorption or gain, timu emission. This kind of recombination is prominent in direct-bandgap r lasers.

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12 a)

Radiative Recombi

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hole in the valence ... of electron-hole recombination, a photon is emitted.
Radiative ... tion of the electron from the conduction band to
the vale ... ses: spontaneous emission, absorption or gain,
and st ... on is prominent in direct-bandgap
semic

Pause

Non-Radiative Recombination

In non-radiative recombination, electron-hole recombination takes place non-radiatively. Non-radiative recombination mechanisms in semiconductors are:

1. **Auger recombination:** In Auger recombination, the energy released during electron-hole recombination is transferred to excite another electron in the same band to higher energy levels. The excited electron achieves thermal equilibrium by dissipating the energy to lattice vibrations or phonons. Auger recombination is predominant non-radiative recombination in long-wavelength semiconductor lasers. As the doping density of the semiconductor increases, the Auger recombination lifetime decreases.
2. **Recombination through defects:** Recombination through defects is a two-step process where electrons recombine with holes through defect energy levels in the bandgap. Defects are introduced in the crystal lattice intentionally or unintentionally. The defects in the semiconductor crystal lattice establish a forbidden region and an electron gets trapped by the energy state in that region. If a hole acquires the same energy state before the electron gets re-emitted into the conduction band, electron-hole recombination takes place. This type of recombination is also called Shockley-Read-Hall (SRH) recombination. This type of recombination is utilized in injection lasers.
3. **Surface recombination:** A strong perturbation of crystal lattice can be called a surface. The surfaces in semiconductor crystal lattices create dangling bonds that are exposed to the ambient. Such exposed surfaces absorb impurities from the ambient and become localized centers of defects. The high concentration of defects in surfaces increases the probability of non-radiative recombination. Cleaved facets in injection lasers are introduced for enhancing recombination in semiconductors through surface recombination.

PART B

(a) Explain different types of recombination mechanisms. (7)

(b) A silicon sample doped with 10^{16}cm^{-3} donors at 300K is optically excited such that the optical generation rate is 10^{20} EHP/(cm^{-3} s^{-1}). Find the separation between Quasi Fermi levels and show the positions of equilibrium d quasi Fermi levels if $\tau_p = \tau_n = 2\ \mu\text{s}$. (7)

Si

$E_{Fn} - E_{Fp}$

Pause

12 b)

$N_D = 10^{16}/cm^3 \approx n_0$

$n_i = 1.5 \times 10^{10}/cm^3$

$n_0 p_0 = n_i^2$

$p_0 = n_i^2/n_0 = \frac{(1.5\times10^{10})^2}{10^{16}} = \underline{\underline{2.25\times10^4/cm^3}}$

$g_{op} = 10^{20}$

$\tau_n = \tau_p = 2\times10^{-6}$

$\delta n = \delta p = 10^{20}\times 2\times10^{-6}$

$= \underline{\underline{2\times10^{14}/cm^3}}$

$n = n_0 + \delta n = 10^{16} + 2\times10^{14} = 1.02\times10^{16}/cm^3$

$P = P_0 + \delta P = 2.25\times10^{4} + 2\times10^{14} = 2\times10^{14}/cm^3$

$n = n_i\, e^{(E_{Fn} - E_i)/KT}$

$E_{Fn} - E_i = KT \ln\left(n/n_i\right) = \underline{\underline{0.3473\,ev}}$

$P = n_i\, e^{(E_i - E_{Fp})/KT}$

$E_i - E_{Fp} = KT \ln\left(P/n_i\right) = \underline{\underline{0.246\,ev}}$

$+$

$E_{Fn} - E_{Fp} = 0.593\,ev$ //

12 b)